

Attention Guided CNN-Transformer Framework for Video Based Driver Drowsiness Detection

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ABSTRACT

Driver drowsiness is a major cause of road accidents, leading to significant safety risks and economic losses. This work proposes an attention-guided CNN-Transformer framework for video-based driver drowsiness detection. The proposed approach combines convolutional neural networks (CNNs) with spatial attention mechanisms for effective feature extraction, along with transformer encoders to capture temporal dependencies across sequential frames.

The model is trained and evaluated using the MRL Eye Dataset for eye-state classification and the NTHU Driver Drowsiness Detection (NTHU-DDD) dataset for sequence-based analysis. Data preprocessing techniques such as image resizing, normalization, and augmentation are applied to improve generalization.

Experimental results demonstrate strong performance, achieving an accuracy of 89.95%, precision of 85.23%, recall of 98.53%, and an F1-score of 91.40%. The high recall indicates the model's effectiveness in detecting drowsy states, which is critical for safety-oriented applications.

The integration of attention mechanisms enhances the model's ability to focus on relevant facial regions, while the transformer module enables robust temporal modeling. Overall, the proposed framework provides a reliable and efficient solution for driver drowsiness detection and highlights the potential of hybrid deep learning architectures in intelligent transportation systems.

Introduction

In the modern era, driver fatigue has become an increasingly prevalent issue due to changing lifestyles and reduced sleep patterns. Studies indicate that individuals today sleep significantly less than previous generations, leading to decreased alertness and cognitive performance. This issue becomes particularly critical in the context of driving, where reduced attention and delayed reaction times can result in severe accidents. Driver drowsiness has been identified as a major contributor to road accidents worldwide, causing significant societal, economic, and human losses.

Various approaches have been proposed to detect driver drowsiness, including physiological measurements, vehicle-based indicators, and image-based analysis. Physiological methods such as EEG signals provide high accuracy but are intrusive and impractical for real-world deployment. Vehicle-based methods rely on driving patterns but often fail to directly capture driver fatigue. In contrast, image and video-based approaches using computer vision offer a non-intrusive and scalable solution.

Recent advancements in deep learning have significantly improved the performance of image-based drowsiness detection systems. Convolutional Neural Networks (CNNs) have demonstrated strong capabilities in extracting spatial features such as eye closure and facial expressions. However, CNNs are limited in capturing temporal dependencies across frames, which are crucial for detecting patterns such as prolonged eye closure or slow blinking.

To address these limitations, this work proposes an attention-guided CNN-Transformer framework for video-based driver drowsiness detection. The proposed system combines CNN-based spatial feature extraction with attention mechanisms to focus on critical facial regions, and transformer encoders to model temporal dependencies across sequential frames. The model is trained and evaluated on the MRL Eye Dataset and the NTHU Driver Drowsiness Detection dataset, enabling robust performance across diverse conditions.

The proposed approach aims to provide an accurate, non-intrusive, and efficient solution for driver monitoring systems, contributing to improved road safety and intelligent transportation systems.

Measurement Type	Definition	Examples	Advantages	Limitations
Biological	Uses physiological signals to detect fatigue	EEG, Heart Rate, Skin Signals	High accuracy	Intrusive and uncomfortable
Image/Video-based	Uses facial features and expressions	Eye closure, yawning	Non-intrusive, cost-effective	Sensitive to lighting conditions
Vehicle-based	Uses driving behavior	Steering patterns, lane deviation	No driver contact required	Indirect measurement
Hybrid	Combines multiple methods	Image + vehicle + biological	High robustness	Complex and costly

Table 1. Overview of driver drowsiness detection methods

Literature Review

Several approaches have been proposed in the literature for driver drowsiness detection, ranging from traditional machine learning methods to advanced deep learning techniques. Early works focused on physiological signals such as electroencephalogram (EEG), heart rate, and respiration patterns to detect fatigue. Although these methods achieve high accuracy, they are intrusive and impractical for real-world applications.

Image and video-based approaches have gained significant attention due to their non-intrusive nature. Techniques based on facial feature analysis, including eye closure, yawning, and head movement, have been widely used. Traditional machine learning models such as Support Vector Machines (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN) have been applied for classification tasks. These models achieved moderate accuracy, typically ranging between 77% and 91%, but often struggle with generalization in complex real-world scenarios.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) have become the dominant approach for drowsiness detection. Models such as VGG16, VGG19, ResNet, and MobileNet have demonstrated strong performance in extracting spatial features from facial images. Several studies report accuracies exceeding 90%, with some reaching up to 97% on benchmark datasets. Ensemble CNN approaches and transfer learning techniques further improve performance by leveraging pre-trained models.

However, CNN-based methods primarily focus on spatial features and often fail to capture temporal dependencies across video frames. This limitation affects their ability to detect gradual fatigue patterns such as prolonged eye closure and slow blinking. To address this issue, hybrid models combining CNNs with recurrent architectures such as Long Short-Term Memory (LSTM) networks have been proposed, achieving improved accuracy in sequential data analysis.

More recently, transformer-based architectures have emerged as a powerful alternative due to their ability to model long-range dependencies using self-attention mechanisms. Vision Transformers (ViT) and Swin Transformers have demonstrated superior performance in capturing global contextual information compared to traditional CNNs. These models provide improved robustness under varying lighting conditions and complex real-world environments.

Despite these advancements, several challenges remain. Many existing models are computationally expensive, lack generalization across datasets, or rely heavily on controlled environments. Additionally, some approaches fail to balance precision and recall effectively, which is critical in safety-critical applications.

In this work, we address these limitations by proposing an attention-guided CNN-Transformer framework that combines the strengths of CNN-based spatial feature extraction with transformer-based temporal modeling. This hybrid approach aims to improve both accuracy and robustness in driver drowsiness detection across diverse real-world conditions.

Proposed Methodology

This work presents a hybrid deep learning framework for driver drowsiness detection that integrates spatial feature extraction with temporal sequence modeling. The proposed system is designed to overcome the limitations of traditional methods by combining Convolutional Neural Networks (CNNs) with attention mechanisms and transformer-based architectures.

Overall Framework

The proposed architecture consists of two major components: (i) spatial feature extraction using a CNN with attention, and (ii) temporal modeling using a transformer encoder. The complete workflow of the system is illustrated in Fig. 1.

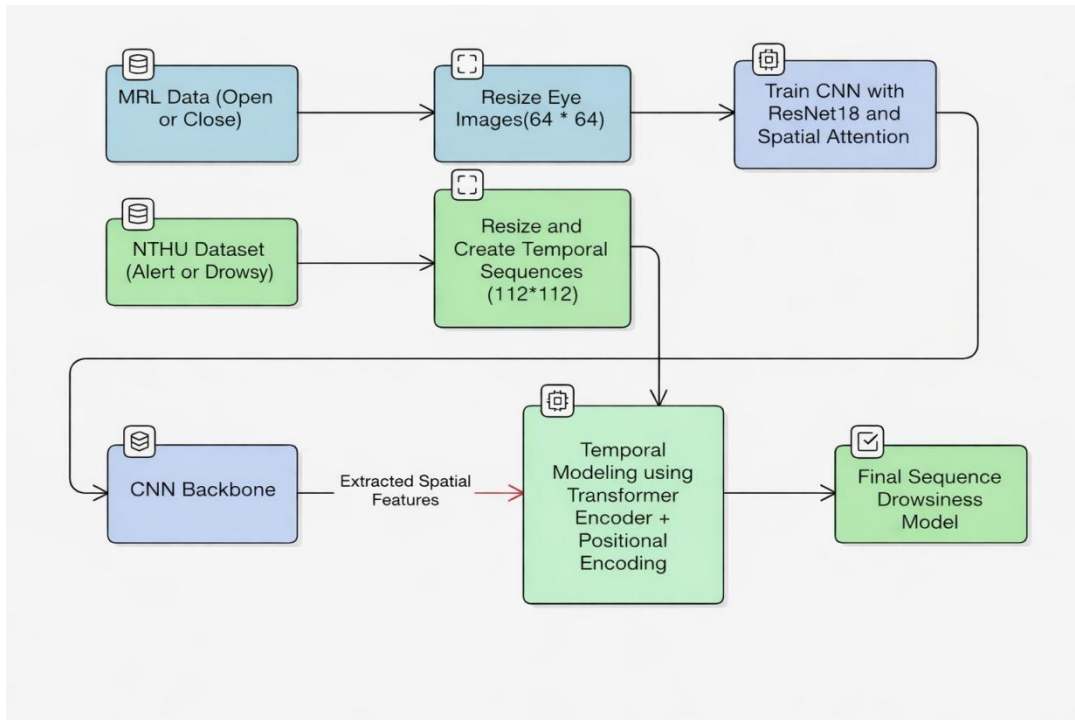


Figure 1. Proposed CNN-Transformer based workflow for driver drowsiness detection.

Dataset Description

The model is trained and evaluated using two publicly available datasets:

MRL Eye Dataset: This dataset contains a large number of eye images categorized into open and closed states. It includes variations in lighting conditions, eye orientation, and image resolution, making it suitable for training robust eye-state classifiers.

NTHU Driver Drowsiness Dataset: This dataset consists of video sequences capturing driver behavior under different conditions such as varying illumination, head pose, yawning, and blinking. It enables the model to learn temporal patterns associated with drowsiness.

Dataset	Key Challenges
MRL	Poor lighting conditions in night mode Presence of eyeglasses Close-up eye images only (no full face) Grayscale images only
NTHU-DDD	Varying facial angles and head pose Different lighting conditions (day/night) Slow blink with nodding Yawning Sleepy combinations Eye occlusion due to glasses

Table 2. Key challenges in drowsiness detection datasets

Data Preprocessing

To ensure consistency and improve model performance, several preprocessing steps are applied:

- Image resizing to fixed dimensions
- Normalization of pixel values
- Data augmentation including flipping and brightness adjustments
- Extraction of eye regions for focused feature learning

Spatial Feature Extraction

A Convolutional Neural Network (CNN) is used to extract spatial features from input images. The CNN is enhanced with a spatial attention mechanism, which enables the model to focus on important regions such as the eyes and surrounding facial features.

The attention mechanism can be represented as:

$$M = \sigma(f([AvgPool(X); MaxPool(X)])) \quad (1)$$

$$X' = M \cdot X \quad (2)$$

where X is the input feature map and M represents the attention map.

Temporal Modeling Using Transformer

To capture temporal dependencies across sequential frames, a transformer encoder is used. The transformer processes sequences of spatial features and models long-range dependencies using self-attention mechanisms.

$$Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V \quad (3)$$

Positional encoding is applied to retain sequence information:

$$PE(pos, 2i) = sin(pos/10000^{(2i/d)}) \quad (4)$$

$$PE(pos, 2i + 1) = cos(pos/10000^{(2i/d)}) \quad (5)$$

Training Strategy

The model is trained in a two-stage process. First, the CNN is trained on the MRL dataset for eye-state classification. The learned features are then used as input to the transformer model, which is trained on sequence data from the NTHU dataset.

Loss Function and Optimization:

The model is trained using Binary Cross Entropy with logits, which is particularly effective for imbalanced binary classification problems. A positive class weighting factor is applied to emphasize the importance of detecting drowsy states:

$$L = BCEWithLogitsLoss(pos_weight = 1.5) \quad (6)$$

The Adam optimizer is used with a learning rate of 1×10^{-4} to ensure stable and efficient convergence during training.

Drowsiness Detection Mechanism

The final system classifies driver states into drowsy or alert categories based on sequential predictions. The model learns patterns such as prolonged eye closure and slow blinking to determine drowsiness.

This hybrid approach enables the system to effectively capture both spatial and temporal characteristics of driver behavior, resulting in improved accuracy and robustness.

Input Representation and Feature Focus

In the proposed framework, the model directly processes input images without relying on traditional region-of-interest (ROI) extraction techniques. Instead of explicitly detecting facial regions or eyes using classical computer vision methods, the system leverages deep learning to automatically learn relevant features.

The convolutional neural network (CNN) backbone extracts spatial features from the entire input image, while the incorporated attention mechanism enables the model to focus on the most informative regions, such as the eyes and surrounding facial areas. This eliminates the need for manual feature engineering or explicit ROI extraction.

By allowing the model to learn feature importance autonomously, the proposed approach improves robustness and adaptability across varying conditions, including changes in lighting, pose, and occlusion.

Ethical Statement

All datasets used in this study are publicly available and commonly used for research purposes in driver drowsiness detection. The data does not contain personally identifiable information and is used strictly for academic and research purposes. All methods and procedures were conducted in accordance with relevant ethical guidelines. No human subjects were directly involved in data collection for this study.

CNN and Transformer Architecture

The proposed framework follows a hybrid architecture that combines convolutional neural networks for spatial feature extraction with transformer-based encoders for temporal modeling.

CNN Feature Extraction: The first stage of the model utilizes a convolutional neural network with a pre-trained backbone to extract high-level spatial features from input images. The network processes images resized to a fixed resolution and generates feature maps that capture important visual patterns such as eye closure, facial structure, and texture variations.

To enhance feature representation, a spatial attention mechanism is integrated into the CNN. This attention module emphasizes the most informative regions of the image, particularly the eye area, while suppressing irrelevant background information. This allows the network to focus on subtle cues such as eyelid movement and partial eye closure, which are critical for drowsiness detection.

The CNN is initially trained on the MRL dataset for binary eye-state classification (open vs closed). During training, transfer learning is applied by freezing lower-level layers and fine-tuning higher-level layers, enabling efficient learning while preserving pre-trained features.

Feature Sequence Construction: After extracting spatial features from individual frames, the feature vectors are arranged into sequential representations. These sequences correspond to temporal frames of driver behavior and are used as input to the transformer model. This step enables the system to move beyond frame-level classification and analyze patterns over time.

Transformer Encoder: The second stage of the model employs a transformer encoder to capture temporal dependencies across sequences of frames. The transformer consists of stacked encoder layers with multi-head self-attention and feed-forward networks.

Unlike traditional sequential models such as LSTMs, the transformer processes the entire sequence simultaneously, allowing it to learn long-range dependencies more effectively. This is particularly useful for detecting gradual drowsiness patterns such as slow blinking, prolonged eye closure, and head movement trends.

Positional encoding is incorporated into the input sequence to retain temporal order information, ensuring that the model understands the sequence structure of frames.

Final Classification: The output of the transformer encoder is passed through fully connected layers to produce the final prediction. The model classifies the input sequence into drowsy or non-drowsy states based on learned temporal patterns.

This hybrid architecture enables the system to leverage the strengths of both CNNs and transformers, where the CNN captures spatial features and the transformer models temporal dependencies. As a result, the proposed model achieves improved performance in detecting driver drowsiness under varying real-world conditions.

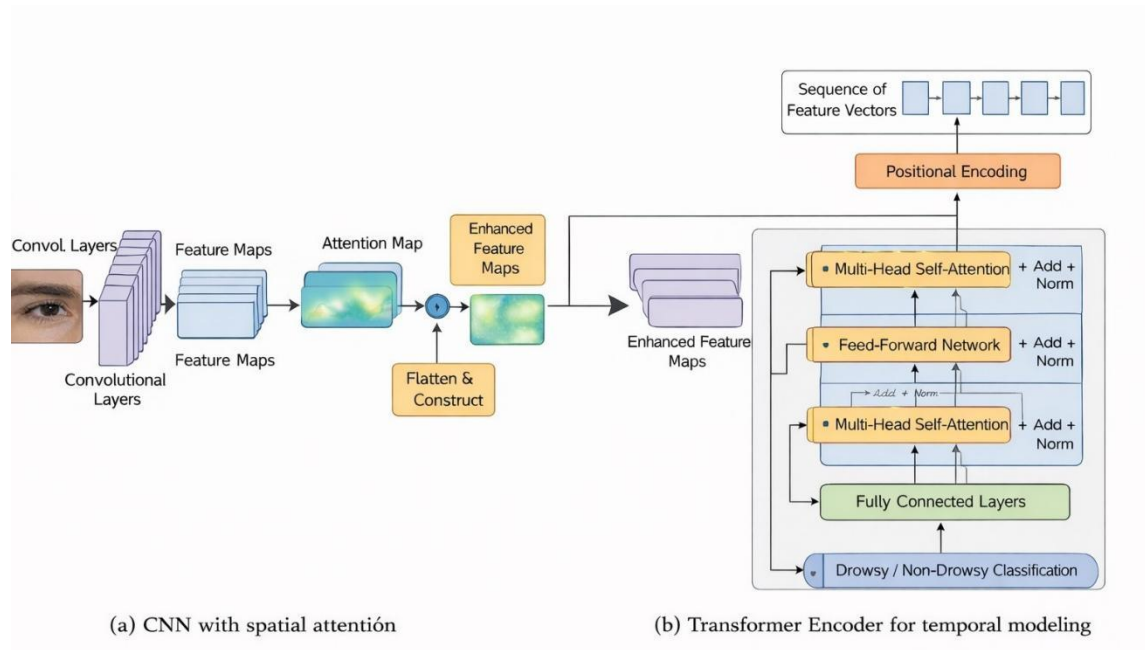


Figure 2. Hybrid CNN-Attention-Transformer architecture for driver drowsiness detection. The CNN extracts spatial features enhanced by attention, while the transformer encoder models temporal dependencies across feature sequences.

Experiments and Results

Experimental Setup

The proposed CNN-Attention-Transformer model was implemented using Python and trained using deep learning frameworks including PyTorch and supporting libraries such as NumPy, Matplotlib, and scikit-learn. The experiments were conducted on a system equipped with GPU acceleration to ensure efficient training.

Component	Configuration
Programming Language	Python
Libraries	PyTorch, NumPy, Matplotlib, scikit-learn
GPU	NVIDIA GPU (CUDA enabled)
RAM	16 GB

Table 3. System configuration used for training and evaluation

Training Procedure

The model was trained using two datasets: the MRL Eye Dataset for eye-state classification and the NTHU Driver Drowsiness Dataset for sequence-based learning. Both datasets were divided into training and testing sets using an 80:20 split.

The CNN component was first trained to classify eye states (open vs closed), and the learned representations were then used to construct feature sequences for the transformer model. The transformer encoder was trained to classify sequences into drowsy and non-drowsy categories.

Hyperparameter Settings

The selection of hyperparameters in this study was based on empirical experimentation and commonly adopted values in deep learning-based image classification tasks. Initial values were chosen based on prior research and were further fine-tuned using a validation split to achieve stable convergence and improved performance.

Hyperparameter tuning plays a critical role in optimizing model performance, particularly for deep architectures involving both convolutional and transformer-based components. The key hyperparameters used in this study are described as follows:

- **Batch Size:** The batch size determines the number of samples processed before updating model parameters. A batch size of 32 was selected as it provides a balance between computational efficiency and stable gradient updates.

- **Learning Rate:** The learning rate controls the step size during optimization. A value of 0.0001 was used to ensure steady convergence without overshooting the optimal solution.
- **Number of Epochs:** The model was trained for a maximum of 20 epochs. This allows sufficient learning while avoiding overfitting.
- **Optimizer:** The Adam optimizer was used due to its adaptive learning rate capabilities and efficient convergence properties.
- **Loss Function:** Binary Cross-Entropy loss was employed as the task involves binary classification (drowsy vs non-drowsy).
- **Early Stopping:** Early stopping with a patience of 5 epochs was applied to prevent overfitting. Training is stopped when validation performance does not improve over consecutive epochs.

These hyperparameter settings ensured stable training, improved generalization, and efficient convergence of the proposed model.

Evaluation Metrics

The performance of the proposed model is evaluated using standard classification metrics, including confusion matrix, accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of the model's ability to correctly classify driver states.

Confusion Matrix: The confusion matrix represents the performance of the classification model by comparing actual and predicted labels. It is defined using four components:

- True Positive (TP): Correct prediction of the drowsy class
- True Negative (TN): Correct prediction of the alert class
- False Positive (FP): Incorrect prediction of drowsy when the actual class is alert
- False Negative (FN): Incorrect prediction of alert when the actual class is drowsy

These values are used to compute the following evaluation metrics:

Accuracy: Accuracy represents the proportion of correctly classified instances among the total number of instances.

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (7)$$

Precision: Precision measures the proportion of correctly predicted positive instances among all predicted positives.

$$Precision = TP / (TP + FP) \quad (8)$$

Recall: Recall measures the proportion of actual positive instances correctly identified by the model.

$$Recall = TP / (TP + FN) \quad (9)$$

F1-Score: The F1-score is the harmonic mean of precision and recall, providing a balance between the two metrics.

$$F1 = (2 * Precision * Recall) / (Precision + Recall) \quad (10)$$

These evaluation metrics ensure a balanced assessment of the model's performance, particularly in detecting drowsy states where minimizing false negatives is critical.

Results

The proposed model achieved the following performance:

- Accuracy: 89.95%
- Precision: 85.23%
- Recall: 98.53%
- F1 Score: 91.40%

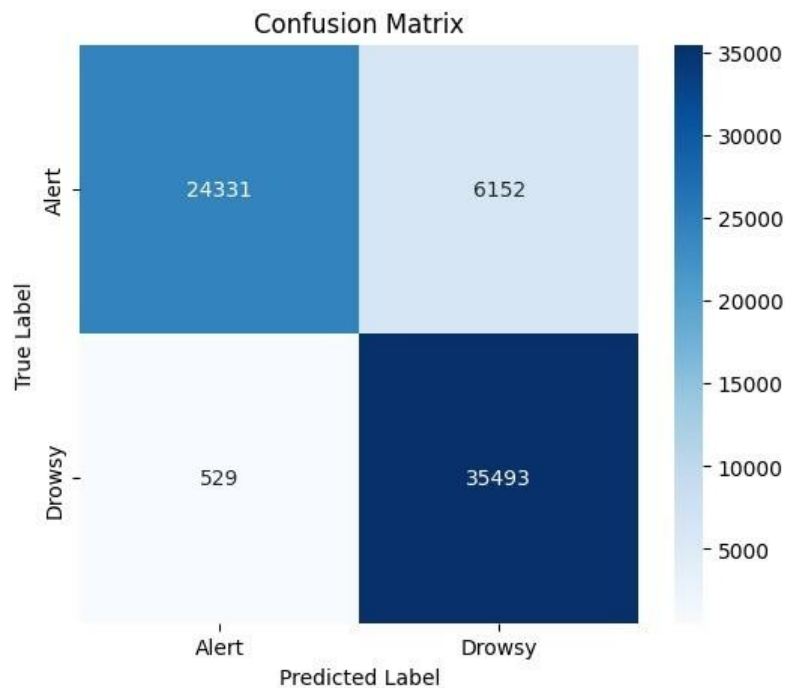


Figure 3. Confusion matrix of the proposed model showing classification performance for drowsy and non-drowsy states.

The high recall indicates that the model is highly effective in detecting drowsy states, minimizing false negatives. This is particularly important in safety-critical applications where missing a drowsy event can lead to severe consequences.

Performance Analysis

The performance of the proposed model is further analyzed using graphical evaluation techniques, including ROC curves, precision-recall curves, prediction probability distribution, and training convergence plots.

ROC Curve:

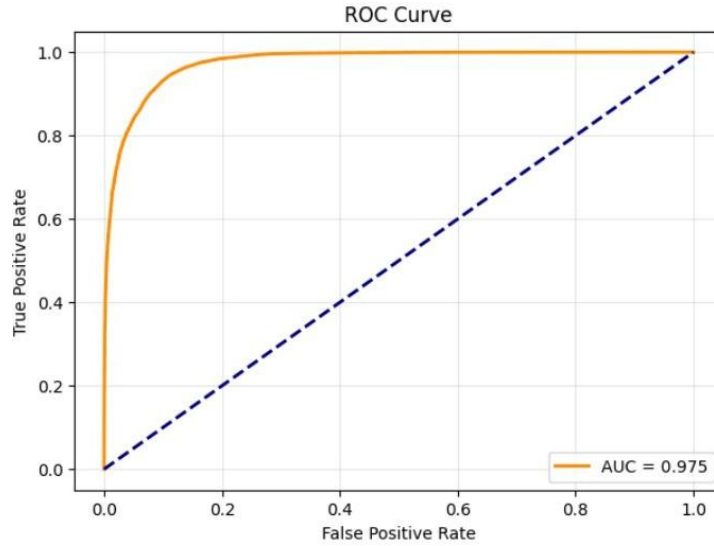


Figure 4. Receiver Operating Characteristic (ROC) curve with an AUC of 0.975, indicating strong discriminative performance of the model between drowsy and non-drowsy states.

The ROC curve demonstrates the model's ability to distinguish between drowsy and non-drowsy states. The high AUC value confirms excellent classification performance.

Precision-Recall Curve:

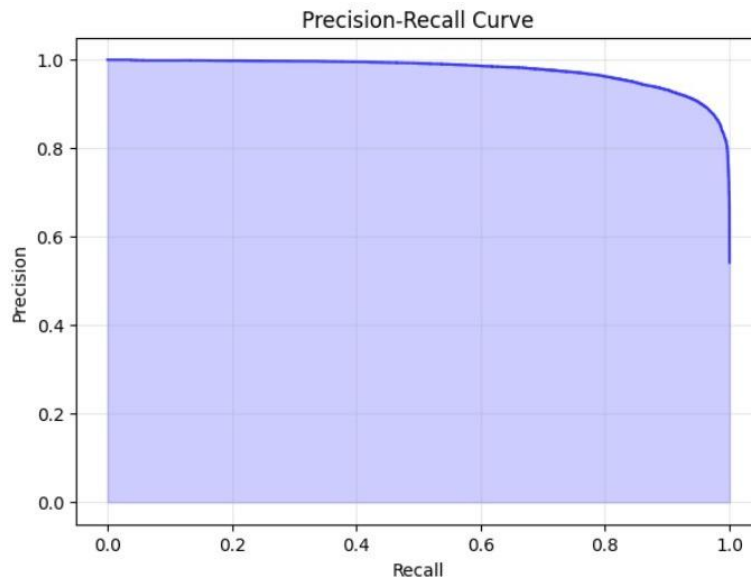


Figure 5. Precision-Recall curve demonstrating the trade-off between precision and recall for the proposed model.

The precision-recall curve highlights the balance between precision and recall. The model achieves high recall, ensuring that most drowsy instances are correctly detected, which is critical for safety-focused applications.

Prediction Probability Distribution:

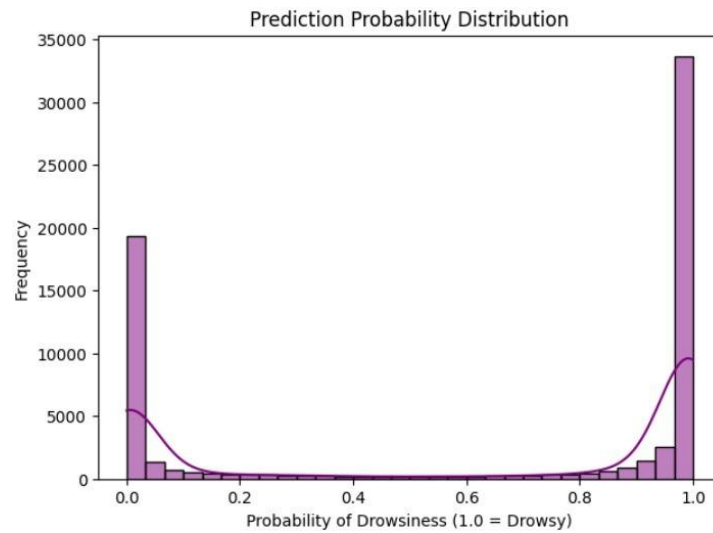


Figure 6. Distribution of predicted probabilities for drowsy and non-drowsy classes, indicating strong confidence in classification.

The probability distribution shows a clear separation between classes, indicating that the model produces confident and well-calibrated predictions.

Training Convergence Analysis:

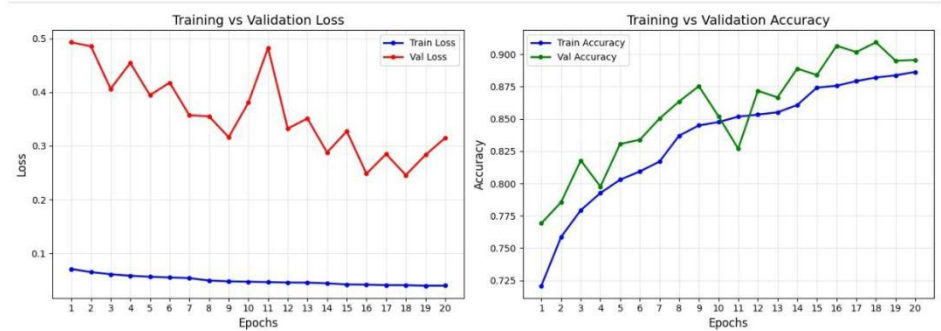


Figure 7. Training and validation loss (left) and accuracy (right) curves showing convergence behavior of the model over epochs.

The convergence curves demonstrate stable training behavior. The training and validation accuracy follow similar trends, while the loss decreases steadily, indicating minimal overfitting and good generalization capability.

Overall, these graphical analyses confirm the robustness and reliability of the proposed CNN-Attention-Transformer model for driver drowsiness detection.

Discussion

The experimental results demonstrate that the proposed CNN-Attention-Transformer model is effective in detecting driver drowsiness with high reliability. The high recall value indicates that the model successfully identifies most drowsy instances, which is critical for safety-oriented applications where missing a drowsy event can lead to severe consequences.

Although the precision is slightly lower, this trade-off is acceptable in practical scenarios, as prioritizing recall ensures that potential drowsiness cases are not overlooked. The confusion matrix further supports this observation, showing a significantly low number of false negatives.

The integration of spatial attention enhances the model's ability to focus on critical regions such as the eyes, improving feature extraction. Additionally, the transformer encoder effectively captures temporal dependencies across sequential frames,

enabling the model to detect gradual patterns such as prolonged eye closure and slow blinking.

The training and validation curves indicate stable convergence with minimal overfitting, demonstrating good generalization capability across datasets.

Conclusion

This paper presents a hybrid CNN-Attention-Transformer framework for driver drowsiness detection. The proposed model successfully combines spatial feature extraction with temporal sequence modeling to improve detection performance.

Experimental results show that the model achieves high accuracy and recall, demonstrating its effectiveness in real-world scenarios. The use of attention mechanisms and transformer-based architectures significantly enhances the model's ability to capture both spatial and temporal features.

Overall, the proposed approach provides a robust, non-intrusive, and efficient solution for driver monitoring systems, contributing to improved road safety.

Future Work

Future work can focus on extending the proposed system to real-time deployment in embedded or mobile environments. Additional improvements may include incorporating multimodal features such as head pose estimation, yawning detection, and facial expression analysis to further enhance detection accuracy.

References

1. Deng, W. and Wu, R., 2019. Real-time driver-drowsiness detection system using facial features. *IEEE Access*, 7, pp.118727-118738.
2. Dipu, M.T.A., Hossain, S.S., Arafat, Y. and Rafiq, F.B., 2021. Real-time driver drowsiness detection using deep learning. *International Journal of Advanced Computer Science and Applications*, 12(7).
3. Jabbar, R., Al-Khalifa, K., Kharbeche, M., Alhajyaseen, W., Jafari, M. and Jiang, S., 2018. Real-time driver drowsiness detection using deep neural networks. *Procedia Computer Science*, 130, pp.400-407.
4. Al Emran, M. et al., 2026. Real-time driver activity detection using advanced deep learning models. *Cognitive Neurodynamics*.
5. Wang, J. et al., 2026. DrowsyDG-Phys: Generalizable driver drowsiness estimation using physiological signals. *Accident Analysis & Prevention*.
6. Hassan, O.F. et al. Real-time driver drowsiness detection using transformer architectures: a novel deep learning approach.
7. Weng, C.H. et al., 2016. Driver Drowsiness Detection via a Hierarchical Temporal Deep Belief Network. *ACCV Workshop*.

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Author contributions statement

Arjun Anand, Aditya Sen, Aryaman Kumar, and Ayush Ranjan contributed equally to the design, implementation, and documentation of this work.

Additional information

The authors declare no competing interests.

Data Availability

The datasets used in this study (MRL Eye Dataset and NTHU Driver Drowsiness Dataset) are publicly available and can be accessed from their respective source